1. **Find first non-repeating character in a string.**
   Code:

```
package Day11;

import java.util.*;

public class FirstNonRepeatingChar {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a string:");
        String str = sc.nextLine();

        Map<Character, Integer> freq = new LinkedHashMap<>();
        for (char c : str.toCharArray()) {
            freq.put(c, freq.getOrDefault(c, 0) + 1);
        }

        char result = '\0';
        for (Map.Entry<Character, Integer> entry : freq.entrySet()) {
            if (entry.getValue() == 1) {
                result = entry.getKey();
                break;
            }
        }

        if (result != '\0') {
            System.out.println("First non-repeating character: " + result);
        } else {
            System.out.println("No non-repeating character found.");
        }
    }
}
```

2. **Remove duplicate characters from string.**
   Code:

```
package Day11;

import java.util.*;

public class RemoveDuplicates {
```

```
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a string:");
        String str = sc.nextLine();

        StringBuilder sb = new StringBuilder();
        Set<Character> seen = new HashSet<>();

        for (char c : str.toCharArray()) {
            if (!seen.contains(c)) {
                seen.add(c);
                sb.append(c);
            }
        }

        System.out.println("String after removing duplicates: " + sb.toString());
    }
}
```

3. **Find element with maximum frequency.**

   Code:

```
package Day11;

import java.util.*;

public class MaxFrequencyElement {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter a string:");
        String str = sc.nextLine();

        Map<Character, Integer> freq = new HashMap<>();
        for (char c : str.toCharArray()) {
            freq.put(c, freq.getOrDefault(c, 0) + 1);
        }

        char maxChar = '\0';
        int maxCount = 0;
        for (Map.Entry<Character, Integer> entry : freq.entrySet()) {
            if (entry.getValue() > maxCount) {
```

```
            maxChar = entry.getKey();
            maxCount = entry.getValue();
        }
    }

    System.out.println("Character with maximum frequency: " + maxChar + "
(appears " + maxCount + " times)");
  }
}
```

4. **Sort a map by keys.**

   Code:

```
package Day11;

import java.util.*;

public class SortMapByKeys {
  public static void main(String[] args) {
    Map<Integer, String> map = new HashMap<>();
    map.put(3, "Three");
    map.put(1, "One");
    map.put(4, "Four");
    map.put(2, "Two");

    System.out.println("Original Map: " + map);

    // TreeMap sorts by keys automatically
    Map<Integer, String> sortedMap = new TreeMap<>(map);

    System.out.println("Sorted Map by Keys: " + sortedMap);
  }
}
```